1. Let the incident wave be $\mathbf{E}_{\text{inc}} = E_0 \boldsymbol{\epsilon} e^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)}$, and let the induced dipole moment be

$$\mathbf{p} = \alpha(\omega) E_0 \boldsymbol{\epsilon} e^{-i\omega t} \tag{1}$$

The amplitude of its radiation-zone electric field is (see Jackson equation (9.19))

$$\mathbf{E}_{\text{sc}} = k^2 \frac{e^{ikr}}{r} (\mathbf{n}\times\mathbf{p})\times\mathbf{n} = k^2 \alpha E_0 \frac{e^{ikr}}{r} (\mathbf{n}\times\boldsymbol{\epsilon})\times\mathbf{n} \tag{2}$$

giving the scattering amplitude $\mathbf{f}(\mathbf{n})$ through $\mathbf{E}_{\text{sc}} = \mathbf{f}(\mathbf{n}) E_0 e^{ikr}/r$,

$$\mathbf{f}(\mathbf{n}) = k^2 \alpha (\mathbf{n}\times\boldsymbol{\epsilon})\times\mathbf{n} = k^2 \alpha [\boldsymbol{\epsilon} - (\mathbf{n}\cdot\boldsymbol{\epsilon})\mathbf{n}] \tag{3}$$

The optical theorem, equation (10.139), gives the total cross section from the forward amplitude projected on the incident polarization,

$$\begin{aligned} \sigma_t &= \frac{4\pi}{k} \operatorname{Im}\left[\boldsymbol{\epsilon}^* \cdot \mathbf{f}(\hat{\mathbf{k}})\right] && \hat{\mathbf{k}}\cdot\boldsymbol{\epsilon} = 0 \\ &= \frac{4\pi}{k} \cdot k^2 \operatorname{Im}\alpha \\ &= \frac{4\pi\omega}{c} \operatorname{Im}\alpha \end{aligned} \tag{4}$$

2. The frequency-integrated cross section is

$$\int_0^\infty \sigma_t(\omega)\, d\omega = \frac{4\pi}{c} \int_0^\infty \omega \operatorname{Im}\alpha(\omega)\, d\omega \tag{5}$$

The integral in (5) is evaluated by the same arguments leading to the Kramers–Kronig relation. Below is a self-contained adaptation of the derivation in this problem's context.

Writing the field and moment as superpositions of frequency components,

$$\mathbf{E}(t) = \int_{-\infty}^{\infty} \mathbf{E}(\omega) e^{-i\omega t} d\omega \qquad \mathbf{p}(t) = \int_{-\infty}^{\infty} \mathbf{p}(\omega) e^{-i\omega t} d\omega \tag{6}$$

the relation $\mathbf{p}(\omega) = \alpha(\omega)\mathbf{E}(\omega)$ becomes, in the time domain, a convolution

$$\mathbf{p}(t) = \int_{-\infty}^{\infty} K(\tau) \mathbf{E}(t-\tau)\, d\tau \qquad K(\tau) \equiv \frac{1}{2\pi} \int_{-\infty}^{\infty} \alpha(\omega) e^{-i\omega\tau} d\omega \tag{7}$$

where $K(\tau)$ is the response kernel, giving the weight of the field a time τ earlier in the present dipole. Causality requires that the dipole cannot respond before the field acts, so

$$K(\tau) = 0 \qquad \text{for } \tau < 0 \tag{8}$$

We can invert (7) and change the lower limit to 0,

$$\alpha(\omega) = \int_0^\infty K(\tau) e^{i\omega\tau} d\tau \tag{9}$$

Obviously for ω with positive imaginary part, the integral (9) converges and defines an analytic function of ω in the upper half-plane. This means all the poles of $\alpha(\omega)$ are in the lower half-plane.

This is consistent with the single bound oscillator with damping constant $\Gamma > 0$ where

$$\alpha(\omega) = \frac{e^2}{m} \left(\frac{1}{\omega_0^2 - \omega^2 - i\omega\Gamma} \right) \tag{10}$$

whose poles are

$$\omega = \pm\sqrt{\omega_0^2 - \frac{\Gamma^2}{4}} - \frac{i\Gamma}{2} \tag{11}$$

which have negative imaginary parts, as expected.

Because $\mathbf{E}(t)$ and $\mathbf{p}(t)$ are real, $K(\tau)$ is real, so

$$\alpha(-\omega) = \alpha^*(\omega) \tag{12}$$

for real ω. Therefore $\mathrm{Re}\,\alpha$ is even and $\mathrm{Im}\,\alpha$ is odd.

At frequencies far above the resonance ω_0 the restoring and damping terms are negligible and the electron responds as if free:

$$m\ddot{\mathbf{x}} = e\mathbf{E}e^{-i\omega t} \qquad \Longrightarrow \qquad \mathbf{x} = -\frac{e\mathbf{E}e^{-i\omega t}}{m\omega^2} \qquad \Longrightarrow \qquad \alpha(\omega) = -\frac{e^2}{m\omega^2} \qquad \text{as } \omega \to \infty \tag{13}$$

so we can claim

$$\mathrm{Re}\,\alpha \to -\frac{e^2}{m\omega^2} \qquad\qquad \text{as } \omega \to \infty \tag{14}$$

Because $\alpha(\omega)$ is analytic in the upper half-plane and vanishes as $|\omega| \to \infty$ there, Cauchy's theorem applied to $\alpha(\omega')/(\omega' - \omega)$ on a contour made of the real axis (indented around $\omega' = \omega$) closed by a large semicircle in the upper half-plane (which contributes nothing) gives

$$\alpha(\omega) = \frac{1}{i\pi}\mathrm{P}\int_{-\infty}^{\infty}\left[\frac{\alpha(\omega')}{\omega' - \omega}\right]d\omega' \tag{15}$$

where P is the principal value. Taking the real part,

$$\begin{aligned} \mathrm{Re}\,\alpha(\omega) &= \frac{1}{\pi}\mathrm{P}\int_{-\infty}^{\infty}\left[\frac{\mathrm{Im}\,\alpha(\omega')}{\omega' - \omega}\right]d\omega' && \mathrm{Im}\,\alpha \text{ odd} \\ &= \frac{2}{\pi}\mathrm{P}\int_{0}^{\infty}\left[\frac{\omega'\,\mathrm{Im}\,\alpha(\omega')}{\omega'^2 - \omega^2}\right]d\omega' \end{aligned} \tag{16}$$

Let $\omega \to \infty$, larger than all frequencies where $\mathrm{Im}\,\alpha$ is appreciable. Then $\omega'^2 - \omega^2 \to -\omega^2$ and

$$\mathrm{Re}\,\alpha(\omega) \to -\frac{2}{\pi\omega^2}\int_0^{\infty}\omega'\,\mathrm{Im}\,\alpha\left(\omega'\right)d\omega' \tag{17}$$

Matching the $1/\omega^2$ coefficient to the free-electron limit (14), we can write

$$\int_0^{\infty}\omega'\,\mathrm{Im}\,\alpha\left(\omega'\right)d\omega' = \frac{\pi e^2}{2m} \tag{18}$$

Finally, we can calculate (5) as

$$\int_0^{\infty}\sigma_t(\omega)\,d\omega = \frac{4\pi}{c}\cdot\frac{\pi e^2}{2m} = \frac{2\pi^2 e^2}{mc} \tag{19}$$